

Seonghwan Park

Ph.D. Candidate, Intelligent Imaging and Vision Systems Lab, DGIST

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BIOGRAPHY

Seonghwan Park is a Ph.D. candidate in the Department of Robotics and Mechatronics Engineering at DGIST, South Korea, where he works with Prof. Inkyu Moon in the Intelligent Imaging and Vision Systems Laboratory. His research builds AI-driven computational bioimaging systems that turn simple, low-cost optical hardware into quantitative, multimodal readouts of living cells.

The core of his work is physics-informed generative modeling — diffusion models and generative adversarial networks constrained by the underlying wave-propagation physics — applied to the ill-posed inverse problems of digital holography. Representative contributions include unsupervised quantitative phase reconstruction and phase unwrapping from single-shot Gabor holograms, and dual-mode platforms that recover quantitative phase together with virtual three-dimensional fluorescence from a single label-free measurement. He has translated these methods into biomedical readouts for stored red blood cell quality, stain-free cancer cell and rare-cell classification, cardiomyocyte drug-response screening, and skeletal muscle phenotyping.

In parallel, he studies learning under severe information constraints — photon-limited imaging and privacy-aware federated learning — and applies generative models to the security analysis of optical and classical cryptosystems. He has authored or co-authored 19 journal articles (11 as first author), holds 3 registered and 9 pending Korean and PCT patents, and received the Best Conference Paper Award at ICEIB 2025 and the DGIST Founder's Day Excellence Award in 2026.

RESEARCH INTERESTS

Computational bioimaging · Digital holography and quantitative phase imaging · Physics-informed generative models (diffusion, GAN) · Label-free and virtual fluorescence microscopy · Self- and unsupervised deep learning · Photon-limited imaging · Privacy-preserving and federated learning

EDUCATION

Ph.D. in Robotics and Mechatronics Engineering, DGIST, Daegu, South Korea Mar. 2019 – Feb. 2027 (expected)

- Integrated M.S./Ph.D. program. Advisor: Prof. Inkyu Moon, Intelligent Imaging and Vision Systems Laboratory
- Dissertation: [thesis title to be added]

B.S., School of Undergraduate Studies, DGIST, Daegu, South Korea Mar. 2014 – Feb. 2019

RESEARCH EXPERIENCE

Researcher, Research Center of Robotics and Mechatronics Engineering, DGIST Jan. 2021 – Dec. 2024

- Developed deep-learning pipelines for holographic and fluorescence cell imaging, and AI-based cryptanalysis, under national research programs.

Graduate Researcher, Intelligent Imaging and Vision Systems Lab, DGIST Mar. 2019 – Present

PUBLICATIONS

19 journal articles (11 as first author), 8 international conference papers, 1 domestic conference paper. Impact factors and JCR ranks are from the 2025 Journal Citation Reports (released June 2026).

Journal Articles — Under Review / In Press

- S. Park**, S. Moon, J. Park, B. Javidi, and I. Moon, "Single-shot physics-informed multimodal reconstruction of quantitative phase and virtual 3D confocal fluorescence from a Gabor hologram", *Laser & Photonics Reviews*, under review (2026) [IF: 9.7] (Q1, top 9% in Optics)
- S. Park**, J. Lee, J. Park, and I. Moon, "BiMODiffusion: Physics-aware unsupervised cyclic diffusion for bidirectional virtual fluorescence and phase reconstruction from single-modality imaging", *IEEE Transactions on Multimedia*, under review (2026) [IF: 9.9] (Q1, top 4% in Computer Science, Information Systems)
- S. Park**, J. Lee, J. Park, and I. Moon, "AI-driven dual-mode phase and label-free fluorescence imaging platform using a single-shot Gabor hologram", *Microsystems & Nanoengineering*, in press (2026) [IF: 11.1] (Q1, ranked 1st in Instruments & Instrumentation)

Peer-Reviewed Journal Articles

- S. Park**, O. Jeong, and I. Moon, "A photon-limited federated learning framework for privacy-aware image classification", *Information Sciences*, 123111 (2026) [IF: 6.8] (Q1 in Computer Science, Information Systems)
- S. Park**, J. Park, Y. Kim, I. Moon, and B. Javidi, "Simple and practical single-shot digital holography based on an unsupervised diffusion model", *Engineering Applications of Artificial Intelligence*, 163, 112970 (2026) [IF: 9.0] (Q1, top 9% in Automation & Control Systems)
- O. Jeong, **S. Park**, and I. Moon, "Scalable neural cryptanalysis of block ciphers in federated attack environments", *Mathematics*, 14(2), 373 (2026) [IF: 2.3] (Q1, top 6% in Mathematics)

13. **S. Park**, J. Lee, J. Park, and I. Moon, "HoloFluoNet: Live cell imaging intelligence based on fused holography and fluorescence for virtual staining, cell segmentation, classification, and viability analysis", *Engineering Applications of Artificial Intelligence*, 161, 112323 (2025) [IF: 9.0] (Q1, top 9% in Automation & Control Systems)
12. **S. Park**, H. An, A. Rehman, and I. Moon, "A holographic sensor-integrated deep learning framework for noninvasive assessment of stored red blood cell quality", *Advanced Sensor Research*, 4(12), e00073 (2025) [IF: 5.0]
11. E. Ahmadzadeh, **S. Park**, Y. Kim, and I. Moon, "Automated phenotypic analysis and classification of drug-treated cardiomyocytes via synergized time-lapse holographic imaging and deep learning", *Computer Methods and Programs in Biomedicine*, 269, 108890 (2025) [IF: 6.4] (Q1 in Computer Science, Interdisciplinary Applications)
10. **S. Park**, M. Y. Kim, J. Jeong, S. Yang, M. S. Kim, and I. Moon, "Quantitative analysis of the dexamethasone side effect on human-derived young and aged skeletal muscle by myotube and nuclei segmentation using deep learning", *Bioinformatics*, 41(1), btae658 (2025) [IF: 5.5] (Q1, top 10% in Mathematical & Computational Biology)
9. **S. Park**, Y. Kim, and I. Moon, "Automated phase reconstruction and super-resolution with deep learning in digital holography", *Optics & Laser Technology*, 176, 111030 (2024) [IF: 5.2] (Q1, top 19% in Optics)
8. A. Rehman, H. An, **S. Park**, and I. Moon, "Automated classification of elliptical cancer cells with stain-free holographic imaging and self-supervised learning", *Optics & Laser Technology*, 174, 110646 (2024) [IF: 5.2] (Q1, top 19% in Optics)
7. **S. Park**, Y. Kim, and I. Moon, "Fast automated quantitative phase reconstruction in digital holography with unsupervised deep learning", *Optics and Lasers in Engineering*, 167, 107624 (2023) [IF: 4.2] (Q2 in Optics)
6. **S. Park**, H. Kim, and I. Moon, "Automated classical cipher emulation attacks via unified unsupervised generative adversarial networks", *Cryptography*, 7(3), 35 (2023) [IF: 2.4] (Q2 in Computer Science, Theory & Methods)
5. K. Jaferzadeh, S. Son, A. Rehman, **S. Park**, and I. Moon, "Automated stain-free holographic image-based phenotypic classification of elliptical cancer cells", *Advanced Photonics Research*, 4(1), 2200043 (2023) [IF: 4.0]
4. E. Ahmadzadeh, K. Jaferzadeh, **S. Park**, S. Son, and I. Moon, "Automated analysis of human cardiomyocyte dynamics with holographic image-based tracking for cardiotoxicity screening", *Biosensors and Bioelectronics*, 195, 113570 (2022) [IF: 11.8] (Q1, top 4% in Chemistry, Analytical)
3. E. Kim, **S. Park**, S. Hwang, I. Moon, and B. Javidi, "Deep learning-based phenotypic assessment of red cell storage lesions for safe transfusions", *IEEE Journal of Biomedical and Health Informatics*, 26(3), 1318–1328 (2022) [IF: 7.7] (Q1, top 9% in Computer Science, Information Systems)
2. **S. Park**, Y. Kim, and I. Moon, "Automated phase unwrapping in digital holography with deep learning", *Biomedical Optics Express*, 12(11), 7064–7081 (2021) [IF: 3.2] (Q2 in Optics)
1. F. Yi, **S. Park**, and I. Moon, "High-throughput label-free cell detection and counting from diffraction patterns with deep fully convolutional neural networks", *Journal of Biomedical Optics*, 26(3), 036001 (2021) [IF: 3.3] (Q2 in Optics)

International Conference Papers and Abstracts

8. T. Bui, **S. Park**, L. El Zahar, and I. Moon, "Cycle-consistent diffusion modeling for bidirectional cryptanalysis of double random phase encoding in federated settings", *Proc. SPIE 14033, Three-Dimensional Imaging, Visualization, and Display 2026*, 81 (2026)
7. H. Woo, **S. Park**, J. Lee, I. Moon, and M. S. Kim, "A lightweight self-supervised deep learning framework for automated detection of circulating tumor cells and cancer-associated fibroblasts", *Cancer Research*, 86(7_Supplement), 119 (2026) [AACR Annual Meeting abstract]
6. **S. Park** and I. Moon, "Automated photon-counted image reconstruction using GANs", *IEEE International Conference on Computation, Big-Data and Engineering (ICCBE)*, New Taipei, Taiwan (2025) — **Best Conference Paper Award**
5. A. D. Martin, O. Jeong, **S. Park**, and I. Moon, "Deep learning-based predictive models over pseudo random number generators", *IEEE International Conference on Computation, Big-Data and Engineering (ICCBE)*, New Taipei, Taiwan (2025)
4. L. El Zahar, **S. Park**, and I. Moon, "Generative adversarial network-based cryptanalysis of double random phase encoding schemes", *IEEE International Conference on Computation, Big-Data and Engineering (ICCBE)*, New Taipei, Taiwan (2025)
3. **S. Park** and I. Moon, "Photon-counting imaging with denoising diffusion models", *Proc. SPIE, Optics + Photonics*, San Diego, CA, USA (2025)
2. L. El Zahar, **S. Park**, and I. Moon, "Double random phase-encoded image reconstruction based on denoising diffusion models", *Proc. SPIE, Optics + Photonics*, San Diego, CA, USA (2025)
1. **S. Park**, J. Jeong, and I. Moon, "Automatic classification of cancer cells using digital holography", *Optica Advanced Photonics Congress — Novel Optical Materials and Applications*, NoTu2C.7 (2023)

Domestic Conference Papers

1. **S. Park** and I. Moon, "Unsupervised GAN-based substitution cipher attack model using imbalanced datasets", *Proceedings of the Korea Computer Congress (KIIE)*, 882–884 (2022) [in Korean]

PATENTS AND SOFTWARE REGISTRATIONS

Registered Patents — Republic of Korea

3. I. Moon, **S. Park**, "Deep learning-based apparatus and method for evaluating red blood cells". Reg. No. 10-2761648, registered 23 Jan. 2025 (filed 10-2022-0101534, 12 Aug. 2022)
2. I. Moon, H. Kim, **S. Park**, O. Jeong, "Apparatus and method for cryptanalysis using generative adversarial networks". Reg. No. 10-2522109, registered 11 Apr. 2023 (filed 10-2020-0144471, 2 Nov. 2020)
1. I. Moon, **S. Park**, Y. Kim, "Deep learning-based apparatus and method for phase unwrapping of digital holographic phase images". Reg. No. 10-2512155, registered 16 Mar. 2023 (filed 10-2020-0085656, 10 Jul. 2020)

Pending Patent Applications — PCT (International)

- 3. I. Moon, **S. Park**, Y. Kim, "Method and system for automated cryptanalysis of block and public-key ciphers based on distributed deep learning". PCT/KR2026/008776, filed 8 Jul. 2026
- 2. I. Moon, **S. Park**, "Cell analysis system and method based on virtual fluorescence generation". PCT/KR2025/004767, filed 8 Apr. 2025
- 1. I. Moon, **S. Park**, "Unsupervised learning-based method and system for phase image reconstruction using Gabor holograms". PCT/KR2025/004263, filed 1 Apr. 2025

Pending Patent Applications — Republic of Korea

- 6. I. Moon, **S. Park**, "Apparatus and method for dual-mode image generation and analysis based on a single-shot Gabor hologram". App. No. 10-2026-0127425, filed 10 Jul. 2026
- 5. I. Moon, J. Park, **S. Park**, Y. Kim, "Apparatus and method for privacy-preserving federated learning using photon-limited images". App. No. 10-2026-0127418, filed 10 Jul. 2026
- 4. I. Moon, **S. Park**, Y. Kim, "Method and system for automated cryptanalysis of block and public-key ciphers based on distributed deep learning". App. No. 10-2025-0139155, filed 25 Sep. 2025
- 3. I. Moon, **S. Park**, "Cell analysis system and method based on virtual fluorescence generation". App. No. 10-2024-0157925, filed 8 Nov. 2024
- 2. I. Moon, **S. Park**, "Unsupervised learning-based system and method for phase image reconstruction using Gabor holograms". App. No. 10-2024-0153232, filed 1 Nov. 2024
- 1. I. Moon, **S. Park**, "Method and system for drug screening of cardiomyocytes using holographic images". App. No. 10-2023-0140106, filed 19 Oct. 2023

Registered Software Copyrights — Republic of Korea

- 5. I. Moon, **S. Park**, "Target rare cell identification program based on bright-field images". Reg. No. C-2026-037293, 30 Jul. 2026
- 4. I. Moon, **S. Park**, "Optical cryptosystem security analysis program based on adversarial generative models". Reg. No. C-2025-043253, 21 Oct. 2025
- 3. I. Moon, **S. Park**, "Photon-counting image reconstruction program based on adversarial generative models". Reg. No. C-2025-043254, 21 Oct. 2025
- 2. I. Moon, **S. Park**, "Live cell analysis program based on virtual fluorescence generation". Reg. No. C-2024-037615, 17 Oct. 2024
- 1. I. Moon, H. Kim, **S. Park**, O. Jeong, "Automated classical cipher analysis program based on generative adversarial networks". Reg. No. C-2020-035557, 15 Oct. 2020

RESEARCH PROJECTS

AI-Driven Multi-parametric Liquid Biopsy Platform for Precision Tumor and Tumor Microenvironment Diagnostics

– Nov. 2025 – Oct. 2028 · Participating researcher — label-free detection and classification of circulating tumor cells and cancer-associated fibroblasts.

Development of Platforms for Analyzing Cellular Phenotypes based on AI and Multi-mode Bio-imaging

– Apr. 2023 – Dec. 2026 · Participating researcher — dual-mode holography/fluorescence imaging and virtual staining for live-cell phenotyping.

Research on AI-based Cryptanalysis and Security Evaluation

– Apr. 2020 – Dec. 2027 · Participating researcher — generative and federated deep learning for classical, block, and optical cryptosystem analysis.

HONORS AND AWARDS

Founder's Day Excellence Award , DGIST	2026
Best Conference Paper Award , ICEIB 2025, New Taipei, Taiwan	2025
– 5th International Conference on Electronic Communications, Internet of Things and Big Data — for "Automated photon-counted image reconstruction using GANs".	

TEACHING

Teaching Assistant , Optical Signal Processing, DGIST	Spring 2020
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TECHNICAL SKILLS

Programming	Python, C/C++, MATLAB, Bash
Deep learning	PyTorch, TensorFlow, CUDA; diffusion models, GANs, transformers, self-/unsupervised learning, federated learning, physics-informed networks, segmentation and classification
Optical imaging	Digital holography (Gabor, off-axis), quantitative phase imaging, fluorescence and confocal microscopy, photon-counting imaging, optical encryption (double random phase encoding)
Experimental	Mammalian cell culture, live-cell time-lapse imaging, fluorescence staining, sample preparation and optical bench alignment for holographic imaging
Systems	Linux server administration, multi-GPU training, Docker, Git
Languages	Korean (native), English (professional working proficiency)